Adam Safron ~ Active Inference Insights 006 ~ Psychedelics, Emotions, Th

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Welcome back everybody to the sixth episode of Active Inference Insights. I am back today
with another fantastic guest, Adam Saffron. Adam is a postdoctoral research fellow at Johns Hopkins
Center for Psychedelic and Consciousness Research. With deep roots in the Active Inference Framework,
Adam's work has explored the nature of preferences and motivation, mindfulness, consciousness,
agency, and psychedelics. In addition to this, he might well be our only guest thus far who has
written a paper titled What is Orgasm? and then lifted 280 kilos for multiple reps in sandals,
might I add. Note, by the way, that these two domains need not be related. Adam,
thank you so much for joining me and welcome to the show. So good. I know I'm really glad you could
join us. And as I said, I mean, I'm always blown away by the diversity of interests and research
topics that our guests bring to the table. So it'd be great because I think we're going to have a
wealth of things to cover. The first thing I wanted to ask is about the body. You write in one of your
papers, bodies represent near ideal initial systems for learning and inference with this prototypical
object and causal system providing basis for further modeling. I wanted to ask, because I've been
interested in what Jakob Howie called the evidential boundary. Where can we kind of circle off the
internal and the external with respect to Markov blankets and active inference? So I wanted to get
your take on whether you thought that, what do you think of the body? Is it a prior that we could say
the mind, the cognitive system, it's incorporating in its sort of predictive schema, or is it itself
an object of inference that the mind, you know, conducts probabilistic reasoning over?
Yes.
To which one?
Hello.
We're on.
So in that particular paper, I'm trying to provide an account of basically how core knowledge
gets loaded into mind. So, you know, we need something like core knowledge, but what's a
biologically plausible account of how we might arrive at it? And so then the idea is you're
basically using the body as basically a way of structuring learning curricula to make
them tractable for a computationally bound system and with limited learning opportunities,
like making the most with the least. And the idea is you create basically the body acts as
a near ideal like training wheel system to figure out where the conditions for inference
are fruitfully constrained. And then you can use that in the structure learning process.
Sure.
And so you can think of it as, so the body would be both in a way an evolutionary prior and that the
specific ways in which it's some, you know, like for instance, like the Rolf Pfeiffer, I think,
speaks really well to this in terms of like the way that the limited shoulder, the range of what the
mobility of the shoulder joint, for instance, already would make your arm go out in front of you

and then like be more likely to grab things and where they're like, well, look right there at the right distance for me to both manipulate them and to observe them. And so those would be the kind of like evolutionary priors you might expect, but it's like a, in the bootstrapping process would be, you know, you could also think of it though, as like core knowledge to get to there.
It would be maybe not necessarily an evolutionary prior, but a very reliably learnable posterior.
So, or an empirical prior. So, you know, I think depending on like, but it might be so reliably learned and so canalized eight ways from Sunday that it might as well be an evolutionary prior, even though there is like this mediation by the learning that takes place, but it's basically in the bag because you did it because nature did it so well. So whether you would call it a prior or a posterior depends on your perspective. And some other people who've kind of talked to the body as a prior, like Micah Allen had a good paper. The body is prior talking about the way in which, for instance, our interoceptive states as particularly stubborn predictions would help to make sure that like the whole like generative model is basically orbiting around allostasis. And so like, so yeah.
Yeah. Does it make, does it make intellectual sense to you to carve a dividing line like an evidential boundary or can we, is it, is it more sort of elegant to just say it's philosophically vague?
Like, so yeah. So, um, so I guess like in that tradition of like the, so like the body as first prior, um, there's also like the work with like, um, uh, Anna Chienitsa and, um, others where it's like us starting like mentalizing homeostasis where it's like you, you, you're, um, part of this learning curriculum is, uh, multi-agent and, uh, coupled like, like we're starting within someone else's bodies. And then even when we're on the outside, we might as well be because we're so dependent. And so like, uh, and within like this, also this, um, um, Rolf, um, Fiferian context or, uh, this roboticist, um, and the overlap with like Rodney Brooks also, um, similar kind of thinking, but the world is its own best model or like emergencies and morpho and, and a lot of the lifting being done by morphological intelligence. Um, you know, even when, you know, you're on the outside of the womb, uh, you're, uh, intensely coupling with others in an extended mind sense. So it's like, uh, it, it seems like you definitely, uh, you can't just say, take it like at the body boundary, then the mind stops, you know, it's, um, yeah.
This is a really interesting avenue. I've been thinking about, um, Clark and Chalmers's 98 extended mind hypothesis paper in the lens of active inference. And I've been sort of informed in my thinking by two people. Um, I've already mentioned one of them on the earlier podcast, which is Manuel De Lander and his most recent book, um, uh, what's it called material phenomenology.
And he gives a sort of, uh, mechanistic account of vision and he critiques the inactivist tradition on this count that basically what they want to call an intrinsic relation, i.e. the coupling itself is constitutive of the identity. So in the coupling between agent and arena, you actually form the definition of agent arena. He criticized it that that, that what actually that really is, is a causal extrinsic relationship. So that's one strand. And then the other is Thomas Metzinger and Thomas, a, what did the podcast of professor Friston. And he spoke about how the notion of inaction might be quite misleading because it kind of implies that there is, there might be a subtle

reivocation of a Cartesian ego where you have a enacting B. So I want, you know, I wanted to kind of pick your brain on that because it's interesting. You brought up extended cognition and the fact that the mind might, its identity might go beyond the, the remits of the body. Are we, are we in danger there of invoking what De Lander calls an intrinsic relation where really what we're looking at is just causal connections? Uh, and we're not, we shouldn't actually be mistaking that for identity relations. I'm not sure. Um, and, and, you know, I might also, um, like in that paper, I imagine I like kind of antagonize more than a few people and actually kind of like going, uh, Neo Cartesian in a way, like suggesting like something like quasi homuncular and you're asked is like, as like essential ways of like understanding like agents and their relations to the world in terms of like the way they're situated and that, um, internalized, having an internalized, uh, representation, uh, pegged according to this, uh, reference frame of an observer, um, from an egocentric perspective, uh, would do a lot of like heavy lifting in the process of structure learning and making things tractable. Um, so, um, the, yes, it's like, I guess, yeah, you might have to like loop around a few times, you know, in terms of getting me to dock properly. But, uh, I guess the way I'm thinking, it's like, like mind as model has to be like extended. And it's like the, um, the ways in which like the affordance landscape is so dependent on these couplings to, to like, to, to go to internalist is in a way, miss the point. Right. But what lets you do anything? What led you to your niche? But I do have sympathies for the internalist account in terms of making room for things like, um, re-presentation so that you can do things like, um, offline learning and imaginative planning. And like, you know, basically like the world is its own best model whenever possible. It's morphological compute, a compute, you know, uh, emergent, um, morpho, uh, you know, from, uh, and the, the, the couplings of the organism as an environment, you know, put, don't just like push your predictions down to the lowest level of like the brain, push them out into the environment. Yeah. Yeah. Yeah. Yeah. Yeah. Well, exactly. I mean, yes, that, that I guess was the, the question I wanted to come to, which is, um, so it's all very nicely interwoven.
You're right. You, you evoke this notion of the kind of quasi homuncular systems. Um, and you mentioned the frame problem, um, which is a really interesting problem to me, which is kind of how can we be, uh, intelligent in an ignorant manner? Uh, how can we zero in on personal information in the world, given the preponderance of sense data? Intelligent in an ignorant manner. I love that. Um, thanks. It's, it's, it's, uh, it's not mine. I'll shout out John Vavakey. I mean, he's the one who's kind of outlined the frame problem to me more than, more than anyone else. You, so I, this is kind of a leading question because I have a sense of where you're gonna, how you're gonna answer it, given what you just said, but can the frame problem be solved by just kind of a single conscious homunculus at the tip of the prefrontal cortex? And if not, how deep do these, or how deep do you imagine that these quasi homuncular or non, all these mindless cognitive agents, as Manuel de Lando would call them, how deep do they go towards the skin? I'm sorry, I know I've gone right into the deep end on these, but why not?
No, no, it's good. Um,

so it seems like
the
extended and active nature
of minds, um, helps to curtail like frame problems and that you're, you're, you're always
like in the world in particular ways. And if like you're coupling with the environment
is creating basically like, is basically like, in a way, creating like a, an enslaving center
manifold for the whole system that you, you, you always have to keep coming back to, to stay on top
of this action, not just the stand top of this action perception cycle. And on top of, um, this
interchange, interchange, um, you know, the, the full scope of relevance is if you're always kind of
like being brought back home and grounded, uh, in this common task of cybernetic control, uh,
that, that seems to like take some of like the, I don't, I don't know if it quite defangs the frame
problem, but it helps to curtail it a bit. Um, the, and so then, you know, but in terms of also like
taking the, the, the range of, um, you know, potential relevant affordances, it's still, you know,
a big space. Um, but still, you know, the idea of like orbiting it around, um,
an agent's model of itself and its interaction with the world. Um,
I don't know how far, um, homuncular, for instance, representation gets you with respect to that, but
in a way it, um, the idea of like basically one of the core things that you're modeling is the
livid body. The idea of the center of your experience, this, um, uh, you know, so, so the
homunculi, I guess we will, so I guess unpack a monkey. So like the, the thing that people are
worried about would be right. A homuncular regress where it's like, there's a little man in the head
that's doing the observing and they've explained things. And so the reason I kind of, um, start to
push on this issue of like trying to resurrect homunculi and things like Cartesian theaters
is the idea is that the, the folk intuition and, and like is coupled to the phenomenology in a way
that actually might reflect the data structures might actually reflect the, um, way the system
is organically reflect the principles of the controller. Right. And so, um, you know, the,
the importance of taking your perception and giving it a coherent perspectival reference frame.
So something, you know, theater-esque in terms of, you know, um, having like the right geometric
relations of the things of the world and their relationship to you. But then also, for instance,
though, the idea of like an observer that helps that watches the screen, um, the sense of, um,
your embodiment as a central data structure, um, being in close, uh, communication, uh, with, uh,
basically, uh, the systems I wish you would do, uh, visual perceptual synthesis because you know,
you're, um, you know, what your body is doing will inform what you're likely to be, what it means,
what's coming in on your retina and, or the visual spatial, the somatospatial should tell you about
the visual spatial and vice versa. And it's interesting that in robotics, these are actually
the primary data structures they use. So like in the collaborations, um, I've done with like Tim Verbalen,
and I was in Katal, like, um, in their latent SLAM systems, one of the primary things for
situating the robot in the world is basically a conjunction of views and poses. It's like,
and, and, and they're almost like one data structure and that it's, you know, one of the best,

you know, where I'm looking, you know, where, where I'm, where my sensors are oriented are gonna be highly informative for what I'm looking at. And also just where my, the way my body is configured, um, I don't think it's quite guessing, this isn't quite brought into their systems, but, you know, the range of relevant affordances, what am I likely to be seeing? Well, your, your, uh, what your body is doing will be highly relevant to that also. And so the mutual information should be very strong, closely coupled. Right. Yeah. It, it, it, it kind of brings me back to the, to the first question, I guess, which is, I mean, I could probably reword that now using the terms you've just said, I mean, I think internal and external and active inference can be considered a, a, a kind of relative misnomer. Um, because if you know, I, I like Maxwell's rewording of it as, uh, tracking and tracked or modeling and modeled. I think that's quite an effective way of reshifting the focus here. So I guess the way of rewording that question is, you know, are we in invoking a homunculus in the Cartesian theater? Are we a risk therefore of kind of thinking that there's a, a genuine ontological homunculus, which is conducting variational Bayesian inference, uh, and, and, and has a model, um, over which it does computations. I think like that rubs a lot of people the wrong way because for a lot of people, it's just dynamical systems. Um, and so, you know, do these kinds of questions actually call into question the more fundamental, um, dichotomy of tracking and tracked and distinct ontologies, distinct entities in the world?

Potentially. Um, and the, so yeah, I, um, maybe I'm asking for too much, but I want to have it all.
I want to have, um, I don't know what you say. Uh, uh, I don't know.
An active couplings in the streets and like,
recall, like it sheets, you know, like, um, the, you know, to the extent possible, like, you know, there's even no need to appeal to like some sort of model representation for much of what I'm describing of this, like mutual contextualization of like these different sensory systems in terms of they're already coordinated just by having a certain situation on the body itself and in the world and the way like the, um, the, the information is cascading in the context of the inactive coupling. And so there's a, there's a sense, you know, the, the, the, the model of the agent is extended into the world. Its mind is extended already there. And a lot of like the, the work I'm talking about, like the homunculus is the body itself. There's no need for like an initial representation of it. Um, I do though suggest we might have though, um, you know, such basically a, uh, a data structure, you would say, or, um, and, and maybe even, um, potentially involving, uh, you know, I, I, I put this forward basically as a kind of prediction that like nothing rides on in the sense of like, uh, uh, if I'm wrong, sure. Uh, but if I'm right, could be interesting. So, but the specific idea is that there might be sub networks of the brain that actually, um, have a geometric organization about them in terms of, you might, you would even have like within cortical hub structures, potentially a, um, geometric correspondences through the manifold activity in those networks and what's being represented. So you, you could even get like homuncular in terms of like the network is almost little person, like on some portions. And the reason you might want to do this is that, um, within geometric deep learning, like if you give a network the same geometry,

essentially, um, not so like a physical network in terms of like, well, I guess it would be there
in the connective eventually, but like, it could be like, just like a dynamic emergent pattern of,
um, effective connectivity. But like, if the, the network though has the same structure as that,
which, which it is, um, representing or modeling or trying to track,
you end up getting potentially orders of magnitude, greater efficiency.
Yeah. That's really cool. That's really cool.
So it's, it's possible that like, there could be in addition to like, most of what we're doing is like
effortless mastery and the voluncular story, you never need representation, you never need
internalized model. And most of it's, and you would want that too. So that's quicker, more efficient,
but you might also on top of that for the sake of things like imaginative planning, things like having
more nuanced control of the sensory motor engagement, you might also have representations of this, um,
and, um, potentially even, uh, uh, shockingly Cartesian accounts, like in terms of highly
centralized and even kind of looking, uh, with a geometry that would, that would be, uh, of a similar
shape to the body itself.
Yeah. That that's fascinating. I mean, these, uh, these strange loops as Hofstadter would call them
are, uh, seemingly never ending the body having loops of itself, the body having loops of the world.
I mean, it comes back to, I think, um, kind of the cybernetic good regulator theorem and, and
like, is that saying, uh, you know, there's this kind of confusion. Is that saying that all good
regulators have a model of the world or are a model of the world? And I think, I think,
I don't know. I mean, I don't know whether you think that's still a live question in active inference.
I mean, to be honest, to be completely candid and yes, I'm a relative layman with a year's experience
of thinking about active inference and not understanding the maths and the physics, but that's okay.
I think there's still a kind of question there, which is, and I, and Carl, to be honest, was quite
neutral on it when I asked him, which is like, what is it? What, what, what are internal states?
Um, yes, internal. So, so it makes sense that internal states have or embody, have embody beliefs
about the external states. I mean, that is at the heart of like a likelihood mapping. So fine.
But is that it? Uh, is that when we're going to give them is these beliefs and what is it to embody
a sub-personal belief? Is it just an instantiate? Like, I think people find it very difficult to
basically move away from the physical mechanics to the statistical mechanics. And I think there needs
to be a bridging mechanism. So I don't know if you had any thoughts just on like the broader way that
a sort of layman audience, including myself, could make those intuitive or those non-intuitive
distances between statistics and physics or the physical material world a bit more intuitive.
Mm-hmm.
It's like we're working with this account of like, you know, blankets of blankets of blankets.
And, um, um, I guess the way I'm interpreting it is like, um, um,
internal, internal states, core screening, external states, and then nested within their internal states,
core screening, external states, so on and so forth. Like a, in a header hierarchy or a heterarchy
of, uh, nested, nested blanketed structures, um, you know, giving you these multiple levels of

resolution that you can unpack as needed. And so I guess the thing I'm wondering is whether like, uh, I, like, I wonder whether a, uh, all, so, so it seems like, you know, none of it works without, like, the inactivist principles, like, without like, the, um, the, the, the structuring of what's happening by intelligent morphologies, or, um, or without just like, that, um, uh, the nature of the coupling of the system in this world, providing like the grounding and the semantics, um, and the interest, uh, none of it gets off the ground. But, but, but the story still is, it can be told just, just at the level of like, the system and its extended phenotype as this one single model. And like, where do you draw this boundary? What's internal? What's external? It's like, it depends on the scale you're thinking about it. Right. Um, and, and so I guess the place where I wonder is we're going to need internalism is when we get to consciousness. Yes. Like, like, so, so like, so I, you know, I guess you might even call me like a kind of like panpsychist of a stripe in psyche in terms of understood as like mind. But then when you get to like basically consciousness in terms of something like, uh, a system for, we might say there is something that it feels like to be that system then. And for that can do things like, um, imagine for instance, in dream, like for, for that and those elements of experience, it seems like we're going to have to at some point invoke, um, enough, uh, uh, a, a, we're going to need a blanket, uh, uh, a, a, a set of attractors that are internal and can, uh, basically break free of the immediate engagement with the world. Right.
That's part of our experience. And so like for people, I think to actually like account for phenomenology to account for like, you know, like a lot of the things that matter to us, you know, eventually getting to, you know, things like agency, even like free will, we're going to need to, at some point, um, go internalist, but like in that process though, you know, it's just, but, uh, still the story for most of evolution, uh, you know, but, but still like if you go in that direction though, and you lose sight of the, the inactivist principles, well then you, you basically find yourself in the same places that I guess you could say like the mainstream, uh, machine learning goes astray sometimes. Right. You think you're going to handle the problem just with scale or raw compute, or, or you don't understand like the kinds of, uh, learning curriculum you would need to bootstrap intelligent systems or how they would need to be deployed to, to, you know, achieve goals in the world. Um, so, uh, yeah. So I, I, again, like, I, I think I ended up like occupying a very awkward position that's like, you know, the, um, radical and activists will be like, you know, you don't need to invoke these things like, um, uh, uh, cognitive, this might be like, you know, oh no, you're, you're leaning way too hard and say that like embodiment is necessary. And you're, and you're just, and, but I think that is the case though. Like it's, it's the, um, I think everyone was kind of right. And just like emphasizing the count of what's going on.
Yeah. Yeah. Yeah. I mean, I, I, I haven't stated my, my personal claim, but I don't disagree. And I, I think that nuance is, is, is always going to be right in some fundamental sense. Um, all theories are wrong. Cause they're just wrong, uh, different, you know, to different degrees.
All metaphors are wrong. Some metaphors are useful.
Right. Exactly. I mean, I, I was wondering, you know, it's funny that it's not funny. It's interesting that you

say that the adaptivist principles are necessary. Um, it'd be great to have like a pure internal, although I can't label people, but like someone like Jakob Howey, who, you know, if he was here, might, again, I'm not putting words in his mouth. So I'd be curious to see what he thinks might ask whether it's, uh, conceivable that a kind of compute only system could do variational Bayesian inference, um, given its priors. I mean, I think, I think another useful distinction actually to, to unpack for our audience in something that I found really, really useful is this distinction between, uh, having a sort of epistemically viable model of the world. So you are actually, you know, it's descriptive. It accounts for the sensations that it's, uh, observing or its sensation is it's receiving and the normatively viable model of the world, i.e. one where you aren't just dispersing into the heat bath, but you can maintain your own sort of auto poetic organization. And I'm wondering whether you think it's conceivable to any degree, whether both sides of that equation could be fulfilled by a kind of sheltered off, super intelligent homunculus.

Sheltered off and it never comes into contact?

Well, I mean, so, so it, it comes into contact, uh, vicariously through the Markov blanket.

So, so there are, I, I'll, I'll permit active and sensory states. Um, but again, vicariously, and has to do this kind of variational Bayesian inference, um, based on its, its model. And we can kind of get into the maths of that a little bit of like where that model comes from in the first place. It might not be quite what I think we're, we're driving at, but like, one thing I would wonder in terms of like, um, like saying like, no, you don't need, um, um, um, yo things like, uh, active learning to be like, Intel to, um, you don't need to like invoke things like extended mind concepts. You don't need to invoke things like niche construction, but they're like in theory, like you could do everything just from like, uh, uh, a smart enough Bayesian observer. Like, I guess, would that be like the complete classroom is like, like, like, uh, well, yes, like almost like has to be the case, but that, that in theory you could do it that the, the inactivist, um, elements that are being appealed to like, if you were going like in theory, if you want to like build like a, like an AGI or something, uh, you wouldn't need to, but then I guess the question is like, if that in theory though, like, oh, in theory, like a three layer neural network can compute anything. Right. Right.

The question, it just might have to be so big. It doesn't fit in like the universe.

And so it's like, if that in theory can't become in practice, given like available compute, available to systems, we would real, we would feasibly be able to create that. It becomes, I think a little bit moot, although still interesting about like that case of like, uh, like, is it a matter of in principle or in practice? So, um,

um, yeah, it's where philosophy has its limits, perhaps, um, where conceivability arguments have their limits. You know, a question that popped up into my mind is bootstrapping up, uh, intelligent, uh, or, you know, um, it kind of in vitro intelligence. Um, I think, I think one question that I've always had is the priors that, um, are encoded in a general model kind of makes sense in this normatively viable sense. So people just think of homeostasis. We have these kinds of, um,

yeah, prior expectations about the states that we expect to be in and we'll act accordingly to,
to confirm that model. I think people might have a bit of a difficulty applying the same logic to, um,
perceptual inference. So perceptual inference, uh, you know, this idea that basically you're inverting
Bayes, you are trying to derive the state from the observation. I guess the question that I have is
if that's a sort of P of X given Y or your probability of the state given the observation,
obviously that's done through some form of the variational Bayes. People might be asking, well,
where does the P of X comes from? Where does your prior of states before observations, um,
I think what might be the D tensor in POMDPs, where does that come from? Like, is that encoded in the
phenotype in the same way that homeostatic, uh, imperatives might be?
Hmm. Or is it the, the D matrix or the, like, you know, how do you-
Yeah, I think it's the D matrix. Again, I'm not a computationist, but I'm using the D matrix.
Your specific initialization within the environment and like getting the ball rolling, the-
Yes, exactly. Exactly.
It's, you know, it's, so it's like, it seems like, like, it's like, we know, like, without
the inductive biases, like, none of it works. The structural learning is just impossible.
Mm-hmm.
You know, with, without the evolutionary priors, without like, some sort of like, you know, intelligent
initialization, I guess the place where, um, I tend to have a, I guess like a heterodox view
and that like, it's like, well, I grant, like, you, we need something like core knowledge. We need
basically the hosier and priors to lean on. Um, I tend to think that, um, like natural selection had
fat fingers when it came to like, like doing the design, uh, like both in terms of like potential
complexity mismatches between inherited information and then like what comes afterwards, but also due
to like, in like, not just like in, uh, yeah, yeah, yeah, yeah, yeah, yeah, yeah, yeah, yeah,
these people will sometimes refer to like the difference and complexity between like the genome
and the connectome, for instance, you know, but obviously, you know, the genome, you know,
the epigenome, and there's a lot more inheritance than that, like just in like the cell, like a lot,
there's a lot of heritable elements that go off. So the question is like, but even then,
there does seem to be some difference in like, you've squeezed through the bottleneck of like
an ovum. And so from this one cell, you reconstituted the rest in this kind of like,
kind of like auto encoding, like you get to pass through a bottleneck here and somehow,
but from this, you're now the, they'll kind of cost reconstitute a, a network of effective,
connect, of functional relations and effective connections, like a full like brains connect them.
And is, you know, is there enough in there to like, I think is the way you might need to.
Um, the way I tend to like, kind of think about it as like, so one heuristic is, um, you know,
the more, so like what, um, kind of knowledge might, what might be needs, what might be the nature of the
representation or just the instantiation of the algebra, then the more complex it is,
the less likely it is that that is in a more direct sense, uh, specified.
Yeah. So it's like, yeah. You know, if we're talking about like, you know, um, so for instance,

like you can think of like, uh, there's like a prior encoded for instance, in like the, the
fraud brain of like the superior colliculus of like potentially of, um, based on sensitivities
to luminance, it might help be like zoom in on eyes potentially like that could be an example of
something or, or, um, the priors about like where you expect light to be coming from in general.
And this like helps to make things trackable. But then let's say you're talking about some of like,
you know, good old fashioned evolutionary psychology, like at the other extreme, like a,
you know, like a cheater detection module. Well, how the heck would you do that? Right.
Yeah. Yeah. Yeah. Yeah. So it's like, how, um, likely is it that you could basically in this
learning curriculum, have these lessons occur and have this knowledge required, uh, early enough.
And the simpler it is, the more likely it could be there early, um, that there's, um, a meaningful
degree of canalization that is actually part of natural selection. Right. Right. Right.
In general, the, uh, uh, uh, uh, claim I've been making is that, uh, uh, in order to pull this off,
um, a lot of it was indirect, even for a lot of the, the core knowledge people will appeal to, um,
that you basically just needed to create a tractable learning curriculum to get there.
And what natural selection was primarily doing was sculpting learning curricula using like, uh, the,
the, the, the body as positioning as a, as a large source of, of leverage on this to help, um, reliably
acquire the knowledge you need, but it's still though acquired, but, but, and how much the adaptation
is able to do that would be depend, would be, um, the, the, the simpler it is, the more likely you could
meaningfully make that a canalization pathway. Yeah. Yeah. That that's awesome. That's a,
that's a great way of looking at it. I, it's funny. I, my undergrad was in linguistics. Um,
and linguistics was the, was the hotbed for all of this stuff. You know, I mean, Chomsky really,
um, Chomsky and Chomsky's Darwin's Darwin problem. I mean, this is like what he was talking about in
the nineties and he had to sort of pare down what universal grammar was because the big question was,
okay, you can account for, you know, perhaps principles and parameters can account for the complexity
and creativity of language learning about how the hell are these hundreds of parameters going to
be encoded in the genome? Right. Like, and like the versions of like Chomsky and thinking where
he's going minimalist type muscle thing like, yeah, recursion. Yeah. But then if it's just like
something like merge, it's like, well, sure. That's, it's very plausible. Right. Right. Right. Right.
Even quickly in evolution, like the ways that's been suggested, you know, and part of like,
what correspond to like these, like phase transitions we see where all of a sudden, like new orders are coming. I'm pretty,
compressed intervals. Yeah. But, but yeah, but if you're going to like, you know, at the other,
let me like things like, I guess, like, uh, I don't know, Pinkerian like, yeah. So then it's like,
uh, oh, I, that I'm not sure biology can do that. Well, I think he may, I think he made that concession.
Um, Chomsky, at least, I'm not sure about Steven Pinker, but Chomsky certainly does. And as you say,
he has the minimus program with merge. Um, but yeah, I think a lot of linguists might argue that he's,
uh, he's kind of abandoned that nativist project. Um, but anyway, we, we need not go in there.
I'll try and get him on the pod actually. Um, sticking on sort of evolution, my background

is not an evolutionary biology or psychology. So I'm, I'm super keen to learn. Um, again, just to take a quote of yours, um, you say much of the functional significance of specific connections in complex neural networks, maybe inherently difficult to predict due to the sensitivity of brackets, chaotic self-organizing systems to initial conditions. So when I hear that, I think, sure, like it certainly makes sense in terms of initial volatility, probably having more of an effect on self-organization than once you have a kind of robust phenotype, but why therefore is there a convergence within a species, a phenotype to kind of a homogeneity, if there is this kind of vulnerability to, um, to chaos?

It seems like a good amount of it. I mean, that, that a good, uh, you know, to get, so you're having a system, which has, um, that's massively recurrent in terms of like, it's like in, in the system as it develops, it feeds back on itself. And then that influences, it feeds back on itself and that influences. And so there's this potential for amplifications to take you in a lot of different directions, but natural selection doesn't want you to go in any given direction. It wants that homogeneity. It wants you to be like, well fit to be, you know, uh, have high fitness. It wants you to be fit to your, uh, the, the niches you're likely to be encountering and to be able to, you know, survive and proliferate your forms and those circumstances. And so, you know, you don't want to be just like open at, you want that homogeneity, but the process, because it feeds back on itself will be, um, it, it should have this potential for, uh, astronomical bifurcation.

So how do you like reign it in? You know, there seems like, and so, you know, you could just say like, you know, well, that's what the inductive biases are. And so, you know, let there be inductive biases and it was good. Right. But what, well, which ones and how, and so I wonder the extent to which like a choke points for, um, coordination among sub elements of the system, such as for instance, like any kind of place you could identify something like a bow tie motif, any place where you can get something like a large sale convergence divergence, which in general you would expect because you need small worldness for the sake of just efficiently getting a system to be integrated or to have a system be integrated at all that, you know, you want to be able to have the different components function together, but if everything's wired, if everything's connected to everything, that's both chaos and there's, there's no, uh, it's very inefficient. So the idea of like these places where things, uh, uh, from broad extent will pass through and then providing basically an opportunity for, um, enslavement, uh, some, so, um, so some of this could just be a network motifs. Like you look at, um, for instance, um, uh, convergence onto like hub structures that have high centrality, but also things like if you take the value signals, like, you know, things like, uh, dopamine or serotonin and you, um, have those also have, uh, um, the conditions of their release correspond to organism scale events, then that could help basically, um, give coordination to the overall system in reliable ways for this task of, uh, helping the organism make it through the world, helping it to, helping the system to function. So it's like these, these, these code dependencies and from, and choke points, these, um, places where you'll get, um, uh, functional convergence.

I, I suspect that's like, um, um, important for keeping that divergence from happening too much. This, uh, uh,

uh,
integration into organismic significance of, with respect to cybernetic control of the agent pursuing
his goals. And if that has perched, if that basically by virtue of whether we're talking about like
value signals that would then, you know, shape connections, um, or, or, or the, the, the structure of the
architecture itself, if that is positioned such that the whole system can be so enslaved, that
should help to like, even though that should basically help these bifurcations in a way be,
uh, not bugs, but features. And that it's certainly, you know, there are sources of like solenoidal flow
and explorations and like, uh, you know, engaging like local pockets of like high temperature search,
but then like you're reigning back in for staying on top of the action perception cycle.
And so to kind of come back to like consciousness, you know, for me, um, it's basically iterative state estimation
with S with the goal being basically, I want to know a rough and ready estimate of the relationship
between the system and the world generated quickly enough that I can meaningfully inform and be informed
by action perception cycles on the timescale of their formation. And so I can nuance that engagement,
provide some extra skillful control and surfing uncertainty. So it's like, um, and so like,
if the primary like task of the system is this inactive coupling, I feel like that helps to provide
that. Um, uh, that's helps to like tame and harness chaos rather than having a thing sort of, uh, bifurcated
in, in, you know, all the like hyper astronomical number of dimensions that could go. Cool. Yeah.
Yeah. There's a lot of that. I mean, yeah, let's, uh, let's do the consciousness thing. Um,
I was going to jump psychedelics, but like, why don't we start even more fundamental consciousness?
Uh, anyone who's watched the first five episodes or four episodes knows that I love consciousness.
Uh, it was kind of my foray into all of this. So what you've given me there is a function of
consciousness. It's guiding action perception loops. It's, it's, it's kind of a likelihood mapping state
estimation. Okay. But as you can imagine, the follow-up question is going to be, why does it
come with its phenomenal characteristics? Um, um, is there something that's intrinsic? Is there something
that's adaptive to the, those phenomenal characteristics, which facilitate, improve,
augment the function that you just laid out?
I think, you know, people could still, you know, like, you know, uh, you haven't pulled up there's
why can't these functions, you know, all take place in the dark. Why must they like feel like anything at all?
I, I don't think that, uh, that concern is removed. I think it loses some of its force.
If the nature of the, uh, data structure is considered as being like specifically the generation of a
visual spatial and somatospatial field of experience, um,
precision weighted or having like with biases rather like different areas of salience and emphasis
based on organismic need. And like, if you're basically taking like basically the, the, the,
thing you are estimating is the sensorium itself. It's sort of like, well, it feels like something
because like emphasis on the fields, like the, like what the data structure is, is specifically
an inversion of a generative model over the sensorium of an embodied embedded agent.
So that's actually what the process is specifically doing. Now you can still say, well,

why couldn't you do the state estimation of the system in the world in the dark?
Exactly. Uh, you know, I, I think it's down objection though. I would, so the claim is that that objection holds more for theories of consciousness that forget we have bodies. So you could still levy it, but I think it's like slightly addressed by making the primary explanandum, the, uh, appearance of a world to a lived body. If that is the primary explanandum and you're specifically saying, how might dynamics, um, compute that, then I think it, it, it takes that on a little bit more though. Why should there be something that feels like? Because what you're specifically trying to explain is the feels itself.
Right. Now it sounds to me quite convergent with Mark Soames is sort of affects driven account of qualia. This being that sort of these fundamental effective states are, um, are sort of guides in a way, to, um, allostatic behavior. So whether that's the resolution of uncertainty, um, information gain, um, testing model parameters, or actually just minimizing variational free energy. See, that for me was one of the first functional explanations of consciousness that really made me think, oh, that's, that's kind of viable because I can't conceive of a feeling without it being felt.
Um, and so what I've been looking for is these kind of, um, it's almost like a synthetic a priori of consciousness. It's almost like a transcendental argument for consciousness. Does anything like circle back on itself so as to make consciousness a prerequisite for the function? And I think where you and Mark are converging here is quite promising, which is that now someone like David, uh, what, someone like 1995, David Chalmers might go, well, you could just have like an offline signal that says things are uncertain and let's resolve that uncertainty. Granted. But then I take quite seriously this, uh, uh, this older phenomenological Heideggerian idea that like we're beings that take a stance on our own existence and that's got to be felt to some degree. Um, so I'm wondering what you think of, of that spiel and perhaps where there's a points of convergence with, with Mark Soames's theories.
I'd say, um, very strong points of convergence, but also divergence. Okay. The convergence is like, you know, he's, um, uh, he, he, he's bringing in life mind continuity. He's bringing in like the necessary interestedness of the system and also like the, the primary nature of, of affect and of saying like, and like you're saying like this synthetic opera, like, like what do we need to even coherently conceive of this thing? We're like, what's foundational. Well, like if you were to doubt it, the whole like game falls apart. Exactly. Yeah. So unless you have like, unless we get the feels in there, we're not getting anywhere. And unless we get like the, um, you know, and so the, the, I guess the thing I would suggest, uh, the, as, as a point of divergence, uh, would be, I wonder if the specific systems. So, so I guess like for, for me, um, I don't necessarily take unconscious feelings to be an oxymoron, but, um, you know, so, uh, we can use words differently, but sometimes I'll, I'll think of like one way of thinking of like emotions and feelings, almost like a very abstract action perception cycle, where it's like the emotions are the global state changes. I don't, I don't think it, I think by the time you get to that little abstraction, the difference between action perception falls apart, but in a way though, like the, the, the, the, the change in organismic mode to be adaptive for a particular class of situations, whatever that is, that seems to be roughly something like emotion would correspond to this. And where the organismic mode could extend beyond the organism

in terms of the emotion and just like contained within it. But the feeling part, uh, potentially being some sort of, um, uh, inference on top of that process, some sort of tracking of that.
Now we can use words in any way, but like I would personally, um, so that's, that's a, that's the way I think of it. And this way I described thinking of in that, um, entropy paper.
And then I wonder where we, whether we can have both unconscious and conscious feelings and emotions.
Yeah.
Like level where like, you know, we go back down to the individual cell and the cell is like changing into like a sporulated mode and maybe that's like it becoming afraid or something, or like extends a pillus to like exchange a chromosome, someone else, you know, or rather that chromosome, uh, uh, chromatid. And that's like, maybe like a lust mode or like it releases, you know, exotoxins or that's like a, like a, like a, like a aggression mode or something like a rage mode. Like, and so like, I like the idea of like, you know, kind of like the work of like Damasio who like, you know, Mark like engaged with and in beautiful ways, like, you know, uh, the, you know, I, I, I love this idea.
I think it's essential for understanding like the actual cybernetic significance of feelings and emotions actually be coherent about them to actually talk about life, mind continuity.
And to recognize this is an older story than even nervous systems in terms of like the, the, um, the, the harmonics or the, the eigen modes of the organ of the organism to be adaptive, having those to be, uh, uh, to give them an overall coherent shape. But, uh, for the specific systems that Mark talks about, um, I wonder whether they, uh, I guess, uh, have what it takes to do the kind of like re-presentation game, whether, um, they're, um, potentially doing their functioning. Um, so he'll, he'll emphasize like the peri-acaductal gray and, and as, you know, being this, um, integration around organism of need and this sort of like the, the way that, um, um, affects in order for them to, in order for these eigen mode shifts to be coherent so that you're, you're not like eating and doing care at the same time, you know, like that would be a bad idea in a lot of circumstances. Like, you know, you don't want, like, you know, mothers to eat their pups, you know, you don't want like this, um, picking and and the, the, the, the, I think, I don't know if he says color coding, but this like categorical breakdown. Yes. I'm totally with him functionally, but it still seems like those could happen in the dark. Right. And so the thing I'm wondering is in order for them to be given something like phenomenality that we experienced with the conscious feelings, I'm wondering whether that requires something like, um, basically a re-representation into cortical circuits. And so he'll, um, and maybe even within, uh, within somatospatial and visospatial modalities. So I make a, uh, somewhat considered busy, would consider to be a bizarre claim that those might actually be all we have and all other modalities are basically, uh, kind of synesthetically cross mapped into those to be given spatial and temporal spatial coherence so that they actually can be modeled and you can do this like state estimation. So the idea is like your interoceptive states might actually be said to be experienced, have to be spatialized as a kind of like texture or as a kind of like, like I feel I, I have to, I have to temporalize and spatialize that information and maybe part of the way you do it is to enfold them into another modality as part of the inference. So, um, but the, uh,

yes, I guess the difference with Mark is I'm wondering whether the, his account is like like, like, like right on, on the functional level, but, um, the systems though, that would actually, but still though, the systems he's emphasizing would be necessary, but not sufficient conditions in order to obtain sufficiency, they need to basically, um, enslave cortical circuits and that those might actually be the constitutive causal, um, the constitutive realizers of experience potentially by virtue of the fact that like, and again, we, I guess we get to like things like small worldness, um, networks that are basically co-located enough that they can converge on attractors and achieve coherence, uh, on the right time scale. The idea of like, if you're like way, let's, so the idea is like, if you're way far away from like, let's say like some sort of rhythmic attractor over, um, uh, uh, is established, that's, um, calculating something like your sensorium, or it's like entailing something like the inversion of your sensorium on the time scales at which that sensorium is inverted. Um, there might be an in and an out in terms of like, who is able to get their message in a time. And it's possible that the systems he's identifying could be, um, on the outside of it in terms of the time scales at which that cause of closure occurs while still being completely necessary for their functioning. I just don't think we know. And so the evidence says, you know, he would then push back and then say like, you know, oh, well, um, the, uh, let's see. Uh, well, no, you know, I, I lesioned the insula and the emotions are still there.
So there's your, and it's like, yes. Um, but, uh,
I guess in a way you might, you might say that, that, that somewhat supports the, the, that quasi synesthetic account I just said, but, um, I basically, I, I find the, uh, lines of evidence that, that are cited to be, um, notable, but none of them are quite convincing to me. So like, so for instance, like, um, uh, uh, like hydrocephalic, uh, children, like, you know, they're still showing these rich emotive displays, but the issue is though, uh, we don't know. Yeah. Like it could be unconscious emotional. Yeah. We can't ask them. We don't know what to sleep from the inside. Or then you might point to like, um, like decorticate rodents showing like, you know, rich, like, you know, adaptive emotional, uh, responses and, and sophisticated behaviors. But once again, we don't know. Um, or you might like drop an electrode and then stimulate these subcortical areas and then you get experience and the person reports to them. But then the problem is, well, the cortex is intact then. And so how do you know if when you drop that electrode, like you stimulate the hypothalamus and then you've got some sort of rage response or a lust response, was it that that was the realizer or is it, as I'm suggesting that that was basically, um, part of, um, you know, it was, uh, causally important for the whole thing to work, but not necessarily constitutive of the process that, um, computes or realizes the quails.
So this is cool. This is cool. Yeah. I feel like there are two things going on and I'd like to sort of see if we can converge them. So one is a kind of structural functional description, which is augmenting, um, or just providing color to Mark's view, which is that we need these kind of, uh, recursive loops back to the cortex. It needs to enslave the, you know, the, these, uh, subcortical structures need to enslave the cortex.

Now someone listening might go, but that's just a further functional structural anatomical explanation.
On the other hand, you invoke kind of proto back to the homunculi, these kind of proto conscious agents.
We've got like, whether that's a cell, a cluster of cells and so on.
Um, and you've spoken as well about how, you know, one of your, your account of consciousness,
you know, can be cast as a convergence of active inference and integrated information theory.
Um, you know, integrated information theory got some, well, it got some bad press earlier this year
for being pseudoscience. Um, on, for those listening, just on audio, I am doing speech marks. Um,
but one of the claims was that it was pandering in some sense to panpsychism.
And perhaps an, uh, acute listener might be thinking, oh, Adam is maybe pandering a touch
to panpsychism with his idea of lustful, uh, neurons. Where do your sympathies lie?
If we take this sort of broader philosophical position, um, where do your sympathies lie with
panpsychism and let's just say more kind of, uh, I think they're called conscious realist positions,
such as that of Donald Hoffman, for example.
I regret saying this, but I think you could call me a panpsychist of sorts still. And that like,
um, of the sense of, um, anything that somehow managed to thing and coming back to the good
regulator theorem on some level had to entail some sort of modeling information, whether it was the
system and it's interaction with the environment was the model or is a model that within it has models.
Um, but has to have, um, some sort of tracking or predictive information, uh, that allows it to
basically reverse, resist the most, uh, temporarily, uh, uh, defy the second law while serving in
everything it does. But basically that, you know, you getting all mixed up and becoming maximally
probable, a maximum entry distribution. Like, how'd you do that? You did something to do that.
Yeah.
Sort of intelligence, some sort of thing, like a Molly, some sort of predictive information had
to have been there for you to pull off something a little bit like Maxwell demon, demon-esque,
you know, to, to do that. And so, and that's, since I, I'd say I'm a panpsychist and that like,
anytime we're pointing to things, anytime there's attractors, um, you might want to say there's
something mind like going on something, a little bit intelligent. Um, but the, um,
um, I think you might even want to say like some psychological kinds, you know, like, like
we're talking about with like talking about like the individual cells, you know, rage or lust,
you know, um, like even those might be like good ways of thinking about them in terms of describing
their cybernetic significance. Like it's a, you know, coherent global change of state with a certain
semantics that's linked to this modeling. And so if you think of, you know, mind as modeling,
and then as modeling with interest in a value-laden way, you know, you might call that sentience then.
But I guess I want to reserve this word consciousness for the, it feels like something.
And I want to reserve that. And there's, there's something that, you know, it feels like from the
inside to be that system. And, you know, so all these systems would have a kind of a dual aspect
parts of them because they would entail an information geometry. And so you can say, it's like, well,
you know, maybe everything that exists, it's just like, that's the way of describing the dual aspect

nature of them. Um, I think you can make a case for that, but I still want though, like, um, basically, um, okay. So in, in, so integrated world modeling theory, the thing I attempted was to, um, take the major theories, um, specifically focusing on integrated information theory and the global workspace theory, and then basically use, you can say probability theory or machine learning or the free energy principle, but this idea of the system modeling its environment probabilistically as now providing the semantics and removing the question begging from these theories in terms of, or some of it, of why there should be something that it's like to be a complex of integrated information or to be the, um, to have fame in the brain and to be within a global workspace. And so why, well, if what that global workspace is doing is it's something like a very efficient architecture, um, for all, or like some, you know, auto-encoding hierarchy, if it's like an arena for Bayesian model selection, allowing for this iterative state estimation, well, then when we have worse, the questions are somewhat less begged or if like integrated information theory is describing, you know, um, uh, the nature of networks that can do interesting forms of compute, it's like, okay. Um, you know, necessary conditions, you know, likely this, you know, integrated information theory, they, they, they have these, these axioms there that they look at that apply to purportedly to all conscious experiences. And then they try to identify these mechanisms, they postulate mechanisms that could realize them. And then the idea is like, if you start out going from these axioms that characterize all experience, you then create a precise and specific mapping to particular mechanisms that would be, that would allow for their realization. Now you can say this system's conscious and that one isn't. And maybe even this system is conscious in that way and not that way. And, you know, I think it's, it's really promising, but, but the issue, um, I take is, um, uh, basically, um, and we like, like losing the expletantum of, uh, our sensorium, our sensory, our sense world, you know, our embodied experience. And so if what that, uh, if what a complex of integrated information is doing is it's basically describing a system set up for a certain type of dynamics within it that could enable a certain type of, um, message passing and, um, an inference and, you know, belief propagation and inference, and specifically with the thing being entailed by such a system would be the, uh, efficient inversion of a sensorium over, uh, inversion of a generative model over the system of embodied about an agent. Now we're getting somewhere. So, so the idea is saying, okay, um, uh, so I guess I'm, I'm sympathetic to both sides and viewing integrated information theory as, um, having some elements which are unsatisfactory. Um,
the, but you know, I, I don't think it was particularly, I don't think what happened was particularly helpful in terms of neuroscience and, you know, uh,
uh, for a variety of reasons, but so, so the, so the basic approach is this. What if you, um, take the existing theories, the major ones, cross-reference their claims, look for points of overlap and try to use basically degree of convergence and the forms of convergence and the coherence of those stories to adjudicate among different claims. And so that was the basic method. And so, um, you could think of, for instance, um, you know, uh, integrated formation theory will talk about, um, potentially there's a posterior hot zone as the substrate of consciousness.

And, you know, for instance, I say that sounds about right. That for instance, like, you know, if basically the posterior, if you take these different sensory hierarchies all coming together into one multimodal hierarchy, and if you can get, um, basically across that system, its ability to, uh, look back on itself in a very powerful way, in a very, um, you know, uh, both integrated, but also differentiated way, then that could be, uh, uh, that, then you could think of that as basically, um, that that could entail the inversion of a, of a generative model over the sensorium of an embodied embedded agent. Maybe that's what's going on. You could also potentially think of though, this complex of integrated information that has both, uh, integration that where the network properties are set up to facilitate both integration and differentiation. That's actually has a lot of correspondences to workspaces, which balance integrated segregated functioning. And so the ideas like that post-year hot zone could be thought of as a fairly global workspace.
Yeah. It's, you know, and, um, with devils and details of exactly the kind of workspace it is.
And, you know, with, um, with me having some specific proposals. So then like, for instance, uh, okay, so the workspace theory, they might come in with the claim, like, no, no, you need the front of the brain. And so the question is, well, what are we talking about exactly? Are we talking about, um, phenomenal consciousness, like the actual generation of your experiences from moment to moment? Or are we talking about your, your knowledge of experience, your, your knowledge of your experience, your awareness of your ability to report on them and contextualize it in different ways?
Well, then you're probably right. The Bowers theory is probably right that you need to get the frontal lobe in the mix to do that. Although at that point though, by the time you're, you're talking about things like that, you might even need more than the frontal lobe. We actually might be in, in like an ex like a extended mind account in terms of like what actually goes into be self-reflective.
Like, you know, we, like we, we, you know, use like the environment to like offload memory.

It's like the way, like we're, so, you know, the, the workspace, the global workspace or play space might not even be skull bound, you know, but still the, so yeah, so the approach I took was basically to say that, um, the different theories were bringing important contributions that all these brilliant people, that these are all brilliant people who've given a lot of thought and there's, have something important to say. Then the question is, are they maybe using words differently? So like in that debate earlier between like integrated information theory, the global workspace theory, you had this recent like adversarial collaboration, so they'll lose. But it's not clear they were using the word consciousness in the same way. Like integrated information theory is talking about experience like a phenomenality. Global workspace theory, it's sometimes it's like illusionist. Sometimes they're, they're, they're more, it's like more of like a higher order of thought type theory. And, and they're oftentimes not necessarily, but the, you know, it's, it's usually they're talking about, you know, um, you'd say like conscious access as opposed to phenomenal consciousness. And so, so then the idea is that if you're, you try to be more like, like, what are the precise claims that people are making and you dig into them, a lot of the conflict might just be people not communicating enough and say like, oh, I meant this by that term and you meant that. Oh. And instead of there being just like one way we use words, it might be that like the different ways we were using words had different trade-offs and maybe like to come together, we have to like add qualifiers. So instead of like, we tried to say this one word means this one thing, maybe we use like, um, we need to use like two or three words to specify which world we're in and what we're talking, what, what our reference are. And then the claim I make is that once you do that, um, a good amount of conflict actually is not as, um, much as it seems like it needs to be like you're actually significantly like these different theories are actually to a, a, a important degree and a useful degree talking about different things. Right. Right. Yeah. Well, you know, maybe there's a psychological account of the fact that scientists like add, you know, they, they might pretend that they believe in collaborative, uh, work, but there is, there's a, there's a tendency or an inclination towards adversarial boundaries, which are often incredibly constraining. So no, it's, it's a good point. And I think it's a good point as well about how much language really shapes. Again, I don't, I don't mean this in some Sapir-Wolf deterministic way, but shapes the way that we view these, um, these models, the way we view these debates. I mean, I think the most, I think for me comes back to, to what you were talking about, like right at the beginning that I could look at a rock and say, well, there's something kind of intelligent going on there. It's tracking some, you know, it's doing some kind of variational Bayesian inference. Well, you know, I think the two most important words for someone coming into the free energy principle depends on the definition of the free energy principle you like, but my two favorite are looks like. So anything that has a mark or blanket and a, uh, you know, a steady state attractor, uh, non-equilibrium steady state attractor will look like it's conducting Bayesian inference. And that look, the, the, the question of what it is doing, the process theory really gets opened up once you go, oh, the free energy principle is just, you know, it might well just be a descriptive tool. And then we talk about active inference, predictive processing, variational message passing. But I think this, this, this looks like notion is often, um, forgotten. And it's important to keep it in the back of our mind that yes, there is something quite miraculous about the coherence of a stone. Um, but I guess every theory theory has its miracles. Um, and the free energy principle is not, it's, it's not devoid of its axioms. Do you think that's a kind of reasonable thing to say?

I think so. And it's like, yeah, and in general, we're trying to do is like, you know, I guess what, uh, minimize the number of like, you know, deus ex machina, as we need to call it, and the number of miracles. And like, but you know, we're going to need to like, you know, like you mentioned, or like synthetic operatory categories, like these things, you, you need some, you need something to build with, you know, or also in terms of like the, what we're talking about with like, uh, evolutionary priors and like inductive bias is like, you need to start somewhere. You need, you need something must be granted, you know, but, but like, but let's try to like do the most we can with the least. And so, um, and, but, but still those things that you, um, are granted, you know, in a way, like there's a sense in which like, you know, they, they might be like, they're the preconditions

for proceeding at all. And those are the miracles we need to, to go on. And so there's like, we end up like, uh, it's like for the case of like, yeah, like a rock and you just say like, you know, it looks, yeah, I had a section I put into like the, like original and, and, and left it in there.
Maybe unfortunately, like they're doing like the, like the IWMT paper. And I, I felt crazy writing it, but you know, and I think like, if I like emphasize this, like looks as if like, it feels less crazy.
And then say the rock is engaging in Bayesian inference to exist, you know, but I guess, you know, there's a, you know, I guess you could say there's a sense in which like, that's how it is.
At that point, I guess you're almost in like a quasi tautology and that like, like, let's just, okay, so here's like the phase space description of the rock. And then it'll, uh, I see, you say, like, if you keep looking at it, um, well, I guess all the rock though, it might be, we need something like, it might not work for the rock. Like it might need the, like, you need something where like that density of the thing, like you like that you can point to it. And like, there's a combination like the solenoidal flow and gradient flow where the thing is like picking itself back together.
Like a drop of oil.
Well, yeah, like a drop of oil that like, like read that, that grabs itself. And then, yeah, yeah, that, yeah.
Yeah. So once you do that, it's that kind of like, I guess that's Carl's like 2019 paper.
Then it's like a sense of like, you can say like, well, that actually is Bayesian inference.
Yeah.
And that there's a, like where you expect to find it. There's a thing like as a probability distribution.
And then if you just look at its action, it's doing Bayesian inference. And then either like, well, is that a tautology? Maybe, but, um, uh, it's a damn interesting ontology, uh, tautology.
And on top. Yeah. Yeah. No, it's, uh, I remember hearing Carl first on Lex Friedman's podcast, talking about a drop of oil, kind of my first proper exposure to free energy. And I just, what? I just couldn't think, I didn't believe that drops of oil or rocks or molecules could be so interesting. And you're right with, you know, um, science, I don't know who said it, but science is about making the familiar unfamiliar. Um, and we're certainly making rocks and drops of oil pretty unfamiliar when we're talking about them conducting variational Bayesian inference.
Yeah. Someone who might be interesting to talk to, he wrote a paper, um, like the world is neural network. And so like, I think he would, um, potentially get to this issue of like, um, is there just like a way in which like, um, you could just make like, like the ontological claim, like the, the rock is engaging. Like he, I, like he would do, I think, uh, uh, uh, a good job of like making a case for that potential.
Okay, cool. I'll, I'll, I'll check out the paper. I mean, yeah, I mean, for me, this, the question that I literally started this podcast with still stands, which is if we take the drop of oil, it's, it's, it's kind of, yes, can be seen as calculating a posterior. Um, but what does that meet, like, what does that mean? Like what is doing the calculation and what's it doing a, like it, what's the, the P of itself given its environment is just really high. I mean, once

you sort of break that down, it just becomes, it seems to me to just return back to a descriptive account rather than a, uh, process theory of what the, what the drop of oil is actually doing. So, you know, I think where a lot of active influence confusion comes from is that we can't, um, we can't, we, we, we struggle conceptually to get away from a Cartesian ego model when we're thinking about how we are and how the world is. So it's, it's tempting perhaps to think that there's a little homunculus within the drop of oil grabbing itself together and maximizing model evidence for itself.
But even I would not say that neither that was, by the way, I'm not saying that either, but it's, um, it might be tempting for some let's, um, let's get to psychedelics because you're at Johns Hopkins, which is the kind of place to be in terms of psychedelic research. I think a useful place to start, not only for myself, but for our audience is people might have a sense of the history of psychedelics in science, um, and the long history and the taboo and the fact that now these experimental studies are taking place and the taboo is lifting. Perhaps you could kind of give a, uh, brief history if possible of the history of psychedelics in science. Um, you don't have to go back to shamanic cultures, but if you feel so inclined, I, I, I shan't stop you. I mean, that actually is a, um, I'm working on a paper right now of like the deep history of psychedelics, like, um, going both back to shamanic cultures, but, but before in terms of the, um, evolution of the, um, signaling systems that they appear to primarily act upon, uh, in particular, the, um, 5ht2a serotonin receptors and... Do the side, but there's a agonist of that receptor. That particular receptor class. Yeah. And, um, so there's, there's a, a paper I just recently, uh, revised and resubmitted, uh, called, uh, um, albus or alter beliefs under psychedelics, which, um, basically provide, uh, this it's providing a, I guess you might say heterodox account of psychedelics, um, with an active inference.
I read the preprint, so, uh, I'm, I'm familiar.
So, so that's, um, so I guess we'll get to that in a bit, but this, uh, uh, that's like where the focus has been, but the current paper would be like, um, why is it that we have psychedelic experiences?
And like, like, what is it? Is there, is there like, so, you know, you might say, um, maybe in a deflationary sense, you could say like, like Pinker would talk about like, uh, music is a cognitive cheesecake where it's like, you know, if you try to look for an adaptive story about music, you could try, but it's just like, no, there isn't really one that's like that meaningful. It's sort of like, you can, the adaptive story of like cheesecake is like, it's a super stimulus that's created because of this, this, and the other, but like, there is no evolutionary account of cheesecake.
It's a late comer to the scene. And so it's a deflationary account of like music would be like, there's no specific music adaptations. It's just like a super stimulus we create for ourselves, you know, uh, because of other things that we need and other aspects of our cognition.
And so with psychedelics, you could say like, you know, these drugs just happen to be like, you know, a super stimulus that's like created, um, you know, maybe by plants to defend themselves against predators in terms of the, you know, you don't want to quite, um, you know, uh, you know, you are to, um, either repel. And so, you know, maybe like, you know, breaking the good regulation, uh, like, like, like making the thing do anomalous inference could be, you know, a good way

of like, um, repelling your, your, your predators. Um, or you might want to potentially entice in terms of, you know, there's like some symbioses that exist in terms of things like that. Was it, um, reindeer? Um, I don't know if this has been established, but like, you know, um, possibly like by consuming things like, um, what is it? Like, was it lichen? I forget what it is, but like, like one guy, like it might like help them to be more like exploratory in certain parts of their, like seasonal cycle. And so then there's like a symbiotic relationships. Um, and you're actually enticing the organism to come in, but the, um, yeah, so actually that's, that's a very live question of like the deep history of psychedelics and the place I'm gonna, I'm trying to go in this upcoming paper is to argue that, um, you know, so the, it seems like the receptor system itself, um, might have evolved, I'm claiming, um, actually in a way. So, so, um, so the count of psychedelics in the free energy principle, I think that's most well known would be called the Rebus model, or relaxed beliefs under psychedelics. And the idea of being that psychedelics are relaxing your Bayesian priors. And so with your prior expectations relaxed, you are now free to think different thoughts and be updated by the world, experience it more profoundly and be changed by more, more profoundly. And, um, so I've been, um, trying to, uh, uh, nuance that account and saying that like, that might be, um, part of what happens as part of like very high degrees of 5H2A agonism, and that you do get something like that, but that might not be the cybernetic significance of the 5H2A system as a control parameter. Um, and that, you know, in many ways, it might actually be exactly the opposite. And so they first evolved with a, uh, gene duplication event around the Cambrian explosion around with the advent of jawed fishes. And so one of the, um, uh, speculations I make is actually, and this might be, um, predator-prey arms races where someone can basically, if you can specifically bite, then this now like adds fuel to, um, a certain fire of like, you know, I can now go up and precisely swim and bite and kill. And now you have an interest in being able to have that strong prediction that you, uh, use to basically, uh, enslave your inaction and achieve that goal. Um, and, and, and the stronger you can get a prediction, the, the, the more, you know, potentially within limits, the, the, the more reliably you'll converge on that goal as opposed to doing something else. But then you might also have an interest on the side of prey of, uh, not being the person bitten. And so this idea of like, uh, turbo charging the organism, um, doubling down on the priors potentially as a way of allowing for specific goal pursuit. That might've been the original story, but where does this like Rebus account come in? Uh, one of the speculations I'm making, um, along with, it's a paper I'm writing in collaboration with Sue Carter, who, uh, studies oxytocin mechanisms. Um, and there's a follow-up paper I'm writing with, um, before the other one was published with, um, Matt Johnson. But, um, is that actually, um, orgasm and birthing might've been the original psychedelic experiences. And that actually, so you're basically relaxing the priors about the bounds of yourself and having greatly elevated plasticity and capacity for updating. Why would you want something like this? Those would be two cases where you'd want it, whether it's bringing basically to basically bringing either, uh, letting, allowing a, a mother to care for the child long enough to, uh, help it to get past these early milestones and also allowing,

potentially, um, uh, mating pairs to stay together long enough to help these hungry children to the next generation.
Shit.
Okay. Let's, let's get into that then. So what would the, the kind, so there's some coupling going on here in terms of this kind of turbo charging. It's psych, so it's in a sense, it's kind of psychologically adaptive to have really relaxed belief over yourself, over kind of your self model, maybe even your autobiographical self model. Um, so as to facilitate, um, the advent of another person into your life. Right. So like people just qualitatively talk about how they now feel like this child is their purpose. They're kind of secondary to this child in their life. So what's the kind of, what's the causal pathway and what's the chronology in this? Is it, is it a genetic adaptation that, um, occurs kind of simultaneously with like modern childbirth is, or whatever that looks like, or is it, does it, um, does it predate it and then get sort of exacted by, um, birthing start, yeah, birthing, or does it come later down the line? What, what, what, I don't know actually how you would examine that, but what are you kind of claiming in that regard?
So the specific, um, the specific mechanistic pathways that were catalyzed in which ways, uh, we got no idea. Sure.
So for instance, like, uh, um, you know, there's, there's all sorts of things that could contribute, like for instance, um, uh, there seems to be, for instance, like the, the 2A receptor, um, you know, in general, it's, it's, it's actually fairly insensitive relative to something like the 1A receptor, but if you take it and expose it to, um, like, uh, reduce like blood acid, like, or rather reduce blood pH or elevated acidity through something like, uh, hyperventilation, like, you know, some kind of things like, um, carbon dioxide or lactic acid accumulating, then it gets potentiated. And so like, for instance, like, you know, during, uh, sexual activity, during, um, uh, uh, birth, there will be a lot of heavy breathing, um, uh, over an extended period. And, you know, is that actually like an accumulation of carbon dioxide? And actually that's part of a mechanism of potential in these receptors. Maybe we have no idea.
Could it involve something like endogenous DMT? Maybe we have no idea. Could it involve it as Sue is arguing, um, uh, oxytocin invasive presidin in different combinations? Quite possibly.
Um, but we have, we're just beginning that now in terms of the specific pathways, but the, in terms of when this might've occurred and then some more things that could have been important, I'm wondering if there were like a couple waves of strong selection, one probably with like the advent of, um, mammalian, uh, reproduction in terms of like, and, and mammalian care.
So, so the, um, you know, once you're now having to, like we're going to earlier, like, don't eat your child care for it, you know, once you're getting to, and, and you've seen a lot of rodent species, it's actually, you know, a little bit, uh, tenuous that process, right? Like, like the runt of the litter often gets eaten, you know, you're not that far from like the atavistic state of kind of like, eh, of, um, so to get this interest, like, uh, was, uh, I think Pat Churchland will actually talk about like an extension of like self grooming to now. So for instance,

you can maybe think of like, actually like relaxing body maps and then making them plastic for a time on the occasion of birthing or like, for instance, like during nursing. And then like, basically, if you then include, um, the pup into the body or the, your offspring to your body maps, like that could get in the count like that off the ground. Um, or then once you like, for instance, the, um, to the degree you get, um, parental investment, you know, but this would then be like a mammalian story and any of those species, there might've been strong selection on like these, like proto psychedelic pathways. And then, you know, for, and then for humans, you know, it would be like all the things coming together in terms of, um, having, uh, you know, the, these big brains, extremely vulnerable, hungry offspring, which you have to care for, for a long time, oftentimes also needing, you know, more than one, because part of it also of the care is providing, um, you know, protection within the social hierarchy, teaching them things, you know, two heads being better than one. And so having the, you know, the bonding of the parents for the child and the parents to each other, like now the selection pressure might be really strong. Um, what was selected, you know, something involving oxytocin potentially in connection with DMT, with other couples, we got no idea. But wow. But then I guess the one more thing that could happen in, um, not just in humans, but maybe particularly in humans is our, um, our ability to like, like, like, what is it? Uh, Jonathan Hyde talks about like the hive mind switch, like our ability to actually like have collective effervescence and to come into like a group, a sense of group identity for the sake of basically, uh, um, uh, doing things according to fashion, including beating out other groups. And so there seems to have been strong selection on that in the human case also. And so the claim would be that these like basically less pressures for intersubjective, uh, coupling, being particularly strong in humans might've been a source of why, for instance, um, uh, of psychedelic experiences, uh, we're probably not the only species to have them. Um, but the idea being that, um, at least the reasons that we know that we are sensitive to them, some of that could be because of that history of selection, because of those challenges of, of, of bonding between mating pairs, between parents and children and potentially between a tribe.

Jonathan Hyde

Yeah, that, that is really quick. Um, it brings two things to mind. Um, one is, I don't know whether you've ever heard of the obstratecal dilemma, um, which is basically this idea that when humans went from being quadrupedal to bipedal, the hips of the mother narrowed. And what was happening is that there was a lot of, uh, stillborn deaths, a lot of childbirth deaths because the hips were narrowing, but the size of the baby's head stayed the same. So a lot of children were dying in, um, childbirth, a lot of mums, mothers were dying in childbirth. And so there was an evolutionary trade-off between, um, early birth and bipediality. And so what actually happens is, is that children start to get born younger. Pregnancy is reduced in time to, to permit the narrowness of the hips that allows for bipediality, which in turn sets the stage for caregiving and the fact that a child can't just pop out of the womb like a gazelle and go running. So that's the one thing that comes to mind.

Jonathan Hyde

I'm going to include that in the next paper.

Jonathan Hyde

Well, give me a shout out. Um, and then the second, and then the second thing that makes me think of is

maybe this care domain, this fundamental need to expand body models, self models,

could be this kind of consciousness, synthetic a priori we were looking at. Maybe there's an

adaptive pressure to feel this kind of selfless love, agape, whatever you want to call it towards

your offspring. So as you know, for evolutionary reasons, so that you are a caring,

uh, altruistic caregiver.

Jonathan Hyde

That's, I think, potentially very

Jonathan Hyde

Jonathan Hyde

I think it's important for like, so this, and the evolution of cognitive modernity, there seems to be a kind of like chicken egg story or slight dilemma where, um, you know, so like Chomsky like, you know, you know, you know,

talks about language for the service of thought, but like, and so I guess that he somewhat sidesteps his dilemma by saying like language is just useful for structuring thought right from the get go.

Um, but like, let's say you, like you aren't quite thinking that ways and like language is about communication. Like at first, like you, or just, you think you need that to get like language use off the ground.

Jonathan Hyde

Well, then the question is like, um, but if you need language use to be like, who are you communicating to? And so like something that could basically, um, uh, scaffold, um, uh, the, so, so in terms of like the, the, um, elaboration of, uh, forms of consciousness and maybe like the, even potentially like the precision and the, and the, uh,

uh, I don't know if you'd say richness, but like, like, like even consciousness, like phenomenal consciousness and its qualities, its ability to, you know, uh, represents like precise things.

Like, it seems like in kind of, it seems like,

if there's like a use to being able to feel your feelings in a way where you can reflect on them in a way where they can meaningfully, uh, uh, impact your behavior.

Um, if, if that is helping to, um, basically provide some of the selection, uh, for certain, I don't know if this thought works, but for certain complexifications of consciousness, that that could potentially help other places where you might have.

Chicken egg problems.

Like it's a source of pre-adaptation.

Like, I guess that's what I'm wondering.

Yeah, exactly.

I think that's exactly the line I was kind of shooting down.

Yeah.

I mean, this is, yeah.

Wonderful.

Wonderful ideas.

I guess another question that comes to mind is when people think about psychedelics, they probably don't think about relaxed Bayesian priors.

They think about tripping for want of a better word.

They think about hallucinations, visions, auditory, um, hallucinations as well.

So in terms of the sort of adaptive self-organizational, um, yeah, uh, functional utility of psychedelics, that all makes sense.

Where on earth do, uh, the source of, uh, magic dwarves or, uh, recurrent patterns or whatever Terrence McKenna was talking about, what, what's going on there?

And, and how does that complexify this story?

Okay.

So, um, so before we get to machine elves, so part of the heterodox psychological view, uh, uh, rather the heterodox active inference view on psychedelics, um, would be that, um, potentially those types of phenomena as hallucinations or some forms of like, you might say, psychotomimetic cognition.

Or like the, the, um, you know, the entertaining of a, of a, of a thought that you would not normally under entertain.

And you might think to be crazy under other circumstances, but that is there live in your mind that you can maybe think of in some ways, it seems to have like almost the shape of a schizophrenic delusion or like a proto psychotic delusion.

Um, so, um, you could potentially argue for these as a relaxation of high level constraining beliefs, allowing those to be free form.

But you could also think of those as particularly strong priors also that you're imposing onto the world and not, uh, rather you're, you're now saying, I'm going to, I think this is there.

This is my expectation.

And I am not going to let that be contract, uh, contradicted by sense data.

And so that's, so there's a interpretation also in terms of a strengthening of belief of those phenomena you mentioned.

And one of the points I'm trying to make in the paper is I don't think we know how to think about this.

And so it's probably some complex, like admixture of relaxed and strengthened beliefs, both directly and indirectly in different combinations.

Um, where I'm, I guess, trying to say that, like, um, rather than ever saying like psychedelics do X, I want to replace that with, um, this.

This psychedelic, uh, this particular compound at, you know, roughly at, you know, this range of doses does this, this kind, this kind of person with this kinds of set and settings.

And the idea of saying that, like, but the idea of like taking inspiration from Rebus is like relaxed beliefs being an essential feature in some ways and looking to adjusting of your priors.

And your mind is this like belief ecosystem as a fruitful way of like contextualizing psychedelic functions, but then saying, um, moving beyond relaxed beliefs.

Talking about like both strength and relaxed and direct and indirect ways, but to come along to a DMT elves, that would be like an example of like, um, you know, so you can say, uh, you know, uh, you can maybe think of that in a relaxed belief story.

And that for instance, uh, you know, elves do not exist.

And now I've like relaxed certain aspects of my ontology.

And now I've like made room for these imaginings to arrive in a vivid form, like, because of like, um, of an indirect, because I relaxed beliefs at this high level of certain core schemas.

They'd like tell me what I should expect in the world.

Now I've made room for psychedelic elves, for instance, or like, you know, entities.

You could also tell a story though, that these are actually, um, kind of coming back to core knowledge, you know, things like, um, uh, um, uh, agenthood or agency or things like, um, but maybe not even core knowledge, like just early lessons in the learning curriculum and ongoing ones of just, we expect agents in the world.

Cause that's the primary thing we're encountering.

That's our primary niche is each other.

And so then basically for the beliefs, so like in a strength in the count, you could say, let's say you're strengthening your beliefs.

Maybe that you're, you're getting a direct strengthening of these prime, these priors for agency detection.

Are they evolutionary priors?

Are they developmental priors?

We have no idea.

You know, in my thinking, they tend to be more developmental, but it's, you know, it's a spectrum of how, you know, where you want to, how, how, how much evolution shaping went into them.

And as, as, as our conversation earlier, um, yeah.

Also, I, I stumbled across, uh, you saying just these words, which really caught my eye archetypes as core priors.

And I thought, oh, that's fantastic.

What a lovely way to put it.

I mean, okay.

So I, this is just, this is really, really cool.

Um, really exciting, I guess.

So, so just to piece together, to scaffold together, the chronology, the causality, the phenomenology, let's just like try and put it all back together.

So if the original psychedelic experience, let's say it was, uh, childbirth or orgasm, how does this end up?

Uh, is it just a, is it just a coincidence that this ends up in plants?

Um, and if it is, is it informative that it was happening in childbirth or, um, orgasm?

Or is that just kind of a, as I was saying, like a coincidental feature of the activation of this receptor?

Um, because you're, you're, you're, the argument that you said there was earlier, which is that it's a kind of defense mechanism or a kind of, uh, attractor towards symbiosis is quite convincing.

I guess we have to fold into all of this, the shamanic cultures, and then the kind of more modern therapeutic lens on, uh, psychedelics.

Is there a way of coalescing those seemingly disparate strands?

So like for plants, like there's the impact that these signaling molecules, these different ones would have on the plant itself.

And then on the other systems that would interact with them.

So for instance, the, um, you have the 1A and the 2A receptors will tend to have, uh, different impacts on cytoskeleton dynamics.

And so let's say you want to do something like a phototaxis, like the, like that actually, it seems will involve some, um, I'm remembering correctly.

I should remember this precisely, but, um, there might even be like, you can think of it like, um, changing the cytoskeleton dynamics to allow for greater fluidity.

You can maybe think of that as like a kind of belief relaxation for morphogenetic inference of the plant.

So it's like, you can, you can think of the plant as like, you know, you know, doing a phototaxis.

And so, no, I'm here now I'm there.

You know, maybe you think of that as like a kind of like, um, uh, updating for the plant itself.

And, and maybe that's even part very speculatively of why like you find it in something like a cactus where like, that's actually kind of a harder thing because you're blockier and like, you know, something like that.

But then the other aspect would be the defense.

And another reason you might have it in something like a cactus is because like, if you're in the middle of the desert and you're juicy, you're extra tasty.

You know, like in terms of like, there, there isn't a lot of game there.

There aren't a lot of contenders in town for things to eat.

There's actually plenty everywhere, but still you're, you're, you're a target and you're stationary, um, by and large.

You know, so then like, you know, the, which predators were they targeting?

You know, it's like, like insects, um, you know, oh, what's happening to the belief dynamics in the insect.

And so, you know, at that level, you know, maybe, you know, it's, it's largely a, uh, um, a rebus account of, of like these 5C2A agonists for the insects, potentially in terms of like changing, uh, their normal behavior.

I'm not sure how you would think of it there, but like the idea of like, you know, someone, you know, nibbles on you, um, it might be good for them to, uh, go away, become more exploratory and, uh, and maybe even, um, mate like crazy.

Like you actually see something like this in like, uh, locusts.

Um, they have, they're colonized by hallucinogenic fungi, which can be part of their mating cycle and is part of this like period of intense mating they undergo.

And so, you know, you know, you, you know, the question is like, why not just poison them?

You know, maybe it's just like, this was like the best poison came up with.

It was good enough.

And it just did the job and it arrived by some means.

And just the form of poisoning is, um, taking the, the model the organism has by which it is adaptive.

You relax that model, but that's not necessarily a good thing for that, that, that, that, you know, that insect, especially, you know, with his body weight, you know, it's eating a lot of it.

And so the idea of that, that insect, you know, if it's, if it's models are really altered and if those models are trying to constrain all the different ways that system can bifurcate.

Like to like be more homogenous as response in adaptive ways, um, you don't come out necessarily ahead by doing that adaptively.

And so that's just the way, um, insects figured out or their plants figured out how to protect themselves.

Um, is, and it, it, it, it's, and the point is the attack.

Uh, but maybe it's not even an attack.

Um, maybe it's like you want, um, you know, it's, you know, it's fine if you nibble on me, just don't nibble too much.

And then, uh, but just go away.

And then, um, and, and, and then if, if we had this nice relationship of you don't bother me too much.

Um, you know, maybe I even like you to be around.

Maybe I even like that I facilitated your mating and I like that your genes are more prevalent in the next generation.

So, yeah, I guess.

And then coming to, you know, shamans.

Well, before, before you do, just, I'm curious about what you think about this.

Cause just like add maybe a technical instance.

I'm curious about what you think from a rebus account in the, in this case, because you mentioned exploratory behavior there or epistemic foraging, which I thought was really an interesting illusion.

Because in both cases you offered there, there's a move towards epistemic behavior, either whether it's mating and, or, you know, go away in an aversive manner or go away.

Um, and basic, like the reindeer example, you know, spread me around basically in a, in a kind of weird way.

Right.

Yeah.

Well, that's yours.

Go forth.

Right.

And what that screams to me is so it just in, in the sense of like the actual computational modeling in POMDP schemes is an emphasis over expected utility would be sort of high precision over your B and C schema.

Um, and, and E as well, your prior, uh, your prize over action, your habits, basically.

And maybe what's happening here again, it's a completely nascent thought that I haven't given much, obviously, uh, you know, fall to is that relax, you're relaxing your prior beliefs about the consequences of your action, thereby making every single possible action policy equal in terms of epistemic utility.

Which means that you then maximum, you do some sort of jamesian maximization of entropy, which makes the best, uh, in terms of just the, um, expected free energy formula makes the best action policy that which maximizes information gain and all, uh, test model parameters.

So I'm wondering whether that speaks to you in terms of a kind of technical unpacking of what might, you know, what might be happening at a very, um, low level here with insects and cacti.

It was sick.

Well, um, with like changing migration patterns, I think it would fit really well with that.

Um, in terms of, uh, like, like think of like human use cases, it seems like it's a mixture.

Like, so for instance, like, um, so like, um, sometimes, you know, specifically would be for, um, you know, breaking out of some state you're caught in.

Like you go to the shaman and this is part of it, but, but oftentimes it would be like with a very specific interest in mind.

Like, um, who, someone cursed my cow, who is it, you know, or, or like, but, or, or sometimes it would be like, um, uh, uh, so sometimes it seems like it is a good amount about like the epistemics in terms of about like trying to maximize learning and breaking you free from like the thing that is usually owning you from within, you know, like your usual way of achieving your goals, keeping you stuck in a certain pattern.

And then it's like, you need to learn something different, but how are you gonna learn something different if you're always doing the same thing again and again?

And so, yeah, I think that fits for a lot of the, like a lot of the healing uses that accompany like, like an ayahuasca ceremony that that seems like a good description, but they're like other use cases.

Like for instance, like in, um, like, um, it's Amazonian tribes.

Like, uh, it's actually like specifically pragmatic and that like they do it to, to get the hunting party together.

Or like they'll, um, uh, they're about to go to war with a specific other tribe.

But then you could also, they'll make a case of like, um, you know, is it that like they're like their goal of what they want to do when they encounter the other tribe?

You want that to be presented in a very vivid form that imagining you want that counterfactual to be hyper precise.

So that becomes actual, or is it, um, more like a matter of like this kind of collective effervescence from earlier and you actually want a rebus account so that before you go into battle, you're basically, you know, all one body coordinating together, fighting, you know, having each other's backs.

Um, um, or both, and you want, you know, an early stage of something rebus like, and then maybe a later stage of something more, uh, involving belief strengthening.

So it's like, sometimes you're letting go of the pragmatics and sometimes you're actually like holding it on to the, the, um, pragmatic values specifically zooming into it.

Um, it seems like there, like at least in the human case, there is no singular account, but in terms of you're saying about like things like, um, uh, like the, the case of like migration patterns and an animal being changed.

Um, that seems like that, that, that, that, that be relaxing the pragmatics and allowing the epistemics to, to a more optimization for that.

That, that, that could be a really important part of that.

And I think.

Cool.

Amazing.

Yeah.

I mean, we haven't got too much time left, but, um, I guess.

Two more questions.

One totally trivial.

The other one, not trivial tools.

I'll start with the non-trivial one.

Um, at Johns Hopkins, you're, you're really at the heart of the current uptick in psychedelic research.

Are you, well, just to frame this, I sense from an external perspective, there was a great deal of excitement.

You know, starting five, six, seven years ago.

And maybe now there's a, um, there's a slight conservatism that's crept in and, and, and saying, okay, the, the psychedelics might not be the panacea that we hoped it might've been seven or eight years ago.

Again, I'm not on the inside.

I don't know, but that's just my instinct.

What are your kind of, uh, predictions, expectations, level of optimism with respect to psychedelic therapy in the next 10, 20, 30 years?

Um, I have some, uh, so if we can avoid the very live possibility of, uh, backlash as, um, uh, adverse outcomes happen inevitably.

And they will be more notable.

It's, you know, like with driverless cars, like if, you know, even if on net they're saving lives, you will note those cases where it just slams into, you know, a person and you'll just be, and that'll change everything.

And so it doesn't take many people to have some really bad experiences.

Um, they even see like with the recent, like, uh, Matthew Perry, like what happened with him.

And, and, and that really is, um, actually really deeply responsible for everyone involved because, um, I don't want to say that, you know, like ketamine is, you know, uh, not a mixed, uh, like it's a powerful medicine.

And, but, uh, and, and that case, like, um, you know, so you see the shape of it.

There's already like people are interpreting this as, oh, ketamine is dangerous.

And already this has caused people to call for more of a, of a pullback.

But he was also on a, a good dose of an opiate, um, probably has some drinking system and was in a hot tub.

So, you know, so it's like, um, you know, did he die of respiratory depression and then drowned?

You know, it's, uh, but so there's things like that could happen.

And, uh, it doesn't take many of those for the psychedelic renaissance to really be like pulled back and try to make potential like life changing, uh, medicines and interventions.

Um, oh, by the way, uh, whenever I'm speaking, it's, um, outside of a Hopkins position, you know, they, and they are, um, so speaking from the outside, um, uh, they have a particular role.

I'd say in, in the psychedelic renaissance of doing largely conservative work within a mainstream medical model and trying to say, can we push forward that?

And, and, and, and, and how can we approach these medicines safely and responsibly?

And so, um, they tend to be, I'd say more conservative in what they're doing and it's, it's a very valuable function.

Um, and, and so, uh, and in general, they, they try to be very careful.

I'm, I'm a little less careful.

Um, so, uh, the, um, some of the things that are, so some of the open questions and challenges we have is, uh, what, um, I guess you say business models or how can we scale psychedelic therapy?

So, so one of the problems with like a, like a psilocybin intervention is that they're amazingly expensive.

And so the question is like, how can we get like broad access to them?

You need some sort of like group therapy and like a different model than the ones we currently use just to make the economics work, to make this like an available one, a widely available intervention.

Um, the other question is like, if you remove some of the, um, expectancy effects you would get from like the places where these interventions were done and like, um, you know, the, you know, having it done at Johns Hopkins, you know, the, like the, um, or the good Friday experiment, you know, and Harvard, there's a certain trust and authority, which is interacting with this.

Um, as part of like leading up to it, that's part of like the expectation that you will be so changed.

And so if you then try to, you know, have clinics opening up everywhere, do you get the same effect?

Because you'll have, you know, the reports, you know, is some of the five most meaningful experiences in a person's life.

A few sessions, your life has changed.

And to what extent can we still get outcomes like that?

Um, as we move into basically scaling and, and having more institutions and more context and, and do so safely.

One of the things I'm actually very excited about is I have some, uh, friends and, um, pharmaceuticals who are basically creating analogs of these psychedelic compounds to try to change their property slightly.

And so one of them is like, for instance, um, uh, uh, shortening their duration of action.

Like, well, why'd you want to do that?

Well, let's say like you could show up for a therapist and you can do basically MDMA therapy and be sober again in two hours.

Mm-hmm.

And so that would both allow you basically to do it, um, more flexibly, do it, uh, and do it more times potentially.

And, you know, and with less risk from the dose, because things like excitotoxicity will be a function of both the dose and the duration of effect.

And so if you do set a ratio of effect, things like, you know, um, metabolic strain and so you would reduce it.

So those advances, I think are gonna be a complete game changer.

Like, uh, the, the short acting powerful compounds.

I think psychiatry will not be the same once those are on the scene and just countless lives will be saved and transformed.

Fantastic.

So, yeah.

And so there's a lot of like, things are very fraught, right?

You know, we could have something like a couple bad outcomes could cause like a chill on the whole thing.

But the promise if we could just make it through this crucible, I think it's unbelievable.

Amazing.

Well, that's exciting.

And then my, uh, my super trivial question, especially even more trivial, given that response is, um, ayahuasca is, uh, famously kind of an enigma, um, because it's your, you may correct me on my, uh, chemist, my biochemistry yet, but isn't it DMT mixed with a plant among the hundreds of thousands of plants that are found or could be found in the Amazon?

What's your Joe Rogan-esque thoughts on how on earth they managed to synthesize ayahuasca?

Hey, you have your, your edible TMT, but that'll be broken down in a hot minute.

And so you need to find basically this monamine oxidase inhibitor to keep it going.

But among all the different things you could have tried, how did they find the right one?

Quite.

Well, the shamans will totally tell you the plants told us.

Ah.

That's how they did it.

That doesn't help.

Um.

So I guess, you know, I'm hoping, I don't know, maybe we should get like Joe Heinrich on the job and like basically some sort of computational model of like the process of like trying out the permutations of like what shamans do normally.

No.

Uh, maybe it's actually tractable in terms of like, if you see just what they're doing, it's like, yeah, they were going to find that given what they were doing in like over, over 200 years,

you would expect that this percent that they found it.

Sure.

So I would expect something kind of like that, or maybe the plants told them.

Or maybe the plants told them.

Yeah, maybe.

Adam, that was, that was amazing.

I'm kind of mind blown.

Um, that was awesome.

Thank you so much.

Right.

I mean, with every guest, I always feel like I learn just infinite sums of new information, which makes this so much fun for me.

Um, but no, I really thank you.

I know you're a busy man.

And so I'm very grateful for you giving me two hours of your time.

Um, it was a pleasure.

We're going to, we're going to keep digging away at these questions.

Um, but yeah, for me and his shoot, yeah, it was, it was an absolute, uh, pleasure on my part.

So thank you again.

Thank you.

All right.